

Normalising Database Design

Practical 6

Group Work

Scenario: Clinic Management System

A small clinic records patient visits in a single table. Each time a patient visits, the clinic records the following information:

- Patient details
- Doctor assigned
- Treatment provided
- Doctor's consulting room

The current table structure is shown below.

PATIENTID	PATIENT NAME	DOCTORID	DOCTOR NAME	DOCTOR ROOM	TREATMENTID	TREATMENT NAME	TREATMENT COST
P101	John Smith	D01	Dr Adams	Room 5	T01	Flu Test	200
P101	John Smith	D01	Dr Adams	Room 5	T02	Medication	150
P101	John Smith	D01	Dr Adams	Room 5	T03	Injection	300
P102	Mary Jones	D02	Dr Brown	Room 8	T01	Flu Test	200
P102	Mary Jones	D02	Dr Brown	Room 8	T04	X-Ray	500
P103	David Lee	D01	Dr Adams	Room 5	T02	Medication	150
P103	David Lee	D01	Dr Adams	Room 5	T05	Blood Test	250
P104	Sarah Khan	D03	Dr Naidoo	Room 3	T06	Check-up	100
P104	Sarah Khan	D03	Dr Naidoo	Room 3	T02	Medication	150
P105	Peter Molefe	D02	Dr Brown	Room 8	T03	Injection	300

Task 1: Convert to First Normal Form (1NF)

Instructions:

- Ensure:
 - No repeating groups
 - Each field contains atomic values
- Identify the Primary Key
- Draw a dependency diagram for the 1NF

Task 2: Convert to Second Normal Form (2NF)

Instructions:

- Identify the composite primary key
- Identify partial dependencies

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- *Decompose the table accordingly*
 - *Draw a dependency diagram for the 2NF*

Task 3: Convert to Third Normal Form (3NF)

Instructions:

- *Identify transitive dependencies*
- *Create new tables where necessary*
- *Draw a dependency diagram for the 3NF*

Task 4: Final Design Present it in 3NF

Instructions:

- *Clearly indicate: Primary Keys (PK) and Foreign Keys (FK)*
- *Implement your final design in SQL Server Management Studio (SSMS)*

****END****